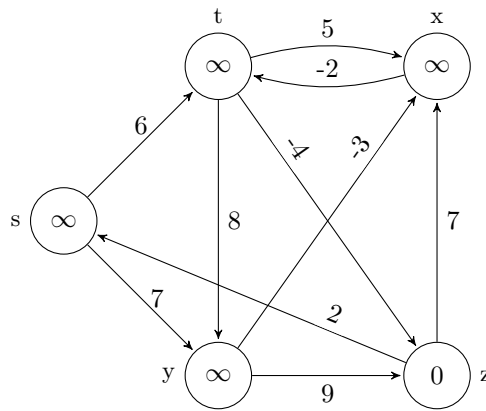
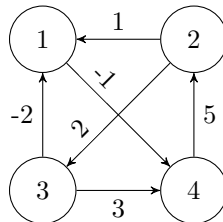


Weekly Assignment 11: Shortest Paths Revisited

- (20 points) Run the Bellman-Ford algorithm on the following directed graph, using vertex z as the source. Show the shortest distance to every state in each iteration.



- (15 points) You are given a directed graph with (possibly negative) weighted edges, in which the shortest path between any two vertices is guaranteed to have at most k edges. Describe how one of the algorithms from the lectures can be modified to find the shortest path between two vertices u and v in $\mathcal{O}(k|E|)$ time.
- (20 points) Run the Floyd-Warshall algorithm on the following graph. Show $D^{(0)}$ and the matrix $D^{(k)}$ after each iteration.



- (15 points) How can we use the output of the Floyd-Warshall algorithm to detect the presence of a cycle with a negative weight? Justify your answer.
- (30 points) Explain how to modify the Floyd-Warshall algorithm to compute the matrices $\pi^{(k)}$ of predecessors. Then write an algorithm **Print-Paths**(π) that prints the shortest paths between all pairs of vertices, one path per line. The path from i to j should be printed as its list of vertices $i \dots j$. If there is no such path, the line should display an error message. Suppose you have procedures **print**(x) that prints x and **newline** that starts a new line.